FastSpeech 2 for English and Telugu

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Study overview

We trained separate English and Telugu models on about five hours of speech per language. We compared character-level and frame-level prediction of pitch, energy and voicing. The frame-level models had lower validation mel error. Pretrained HiFi-GAN also made reference-mel playback clearer, but some generated speech remained noisy.

Language	Best frame-level validation MAE	Completed training
English	0.794563 at epoch 48	50 epochs
Telugu	0.534438 at epoch 56	60 epochs

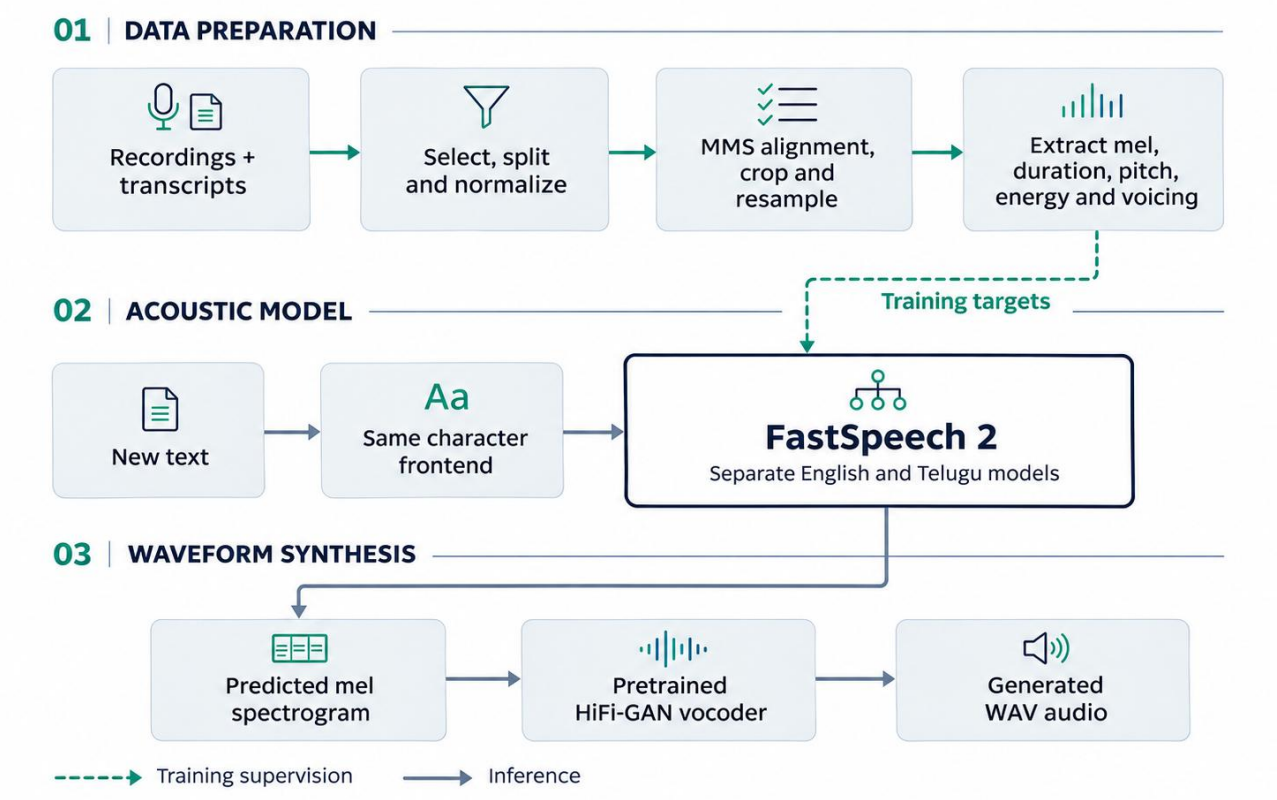


Figure 1. The complete pipeline. Dashed arrows supply training targets; solid arrows show speech generation.

Evaluation scope. Validation mel MAE was measured using recorded durations and prosody. It measures reference-conditioned acoustic prediction rather than text-only intelligibility.

Data and preparation

The two languages were processed separately. We kept the source shard and row ID for every selected clip so that the same examples could be traced throughout the experiments.

Language and source	Training clips	Validation clips	Train / val hours
English · Svarah [5]	3214	630	5.0056 / 1.0014
Telugu · IndicTTS [6]	1697	351	5.0039 / 1.0016

Hours are measured on the centered mel grid, so they are slightly larger than exact cropped-waveform durations. Train and validation have no overlapping source records. Speaker-independent separation is not established.

Dataset preparation procedure

1. Mono recordings of 2–20 seconds were retained. Transcript duplicates were checked using NFKC normalization, case folding and alphanumeric text; audio duplicates were checked using SHA-256.
2. Candidates were shuffled with seed 42. Approximately one hour was selected for validation, followed by five hours for training. Reserve clips were used to reach the duration budget after alignment and cropping.
3. English text was lowercased, and Telugu was romanized with uroman (tel). Internal apostrophes were retained. Characters were concatenated without space tokens. Digit-containing training transcripts were excluded from alignment.
4. MMS supplied character alignments at 16 kHz. Audio was cropped to the first and last aligned characters. Midpoint boundaries were converted to integer mel durations, with at least one frame retained per character.
5. Mel, pitch, energy and voicing targets were extracted. Duration sums were checked against mel length. This established grid consistency without establishing the accuracy of every sound boundary.

Acoustic features and alignment checks

Feature	Exact setting
Waveform and mel grid	22050 Hz; hop 256 samples ≈ 11.61 ms; [frames, 80]
STFT and mel	FFT/window 1024; Hann; center=True; reflect padding; Slaney mel 0–8000 Hz; natural log magnitude floored at 1e-5
Pitch and voicing	TorchCREPE tiny; 16 kHz; 10 ms hop; 65–500 Hz; periodicity ≥ 0.25; −60 dB silence rule
Energy	L2 norm of STFT magnitude across frequency bins
Normalization and vocabulary	Training-only voiced-character pitch and character-energy statistics; reused for frame targets. Padding ID 0, unknown ID 1; English vocabulary 29, Telugu 23.

Pitch is sampled at the nearest mel-frame time. The baseline averages pitch over voiced frames and energy over all frames in each character span. Unvoiced pitch stays zero. Telugu retained the selected female recordings (gender code 0); this does not itself prove a single speaker.

Dataset compatibility. The English loader reads audio_filepath.bytes; Telugu uses audio.bytes. The loader checks the embedded schema rather than assuming a shared column name.

Assessment of phoneme alignment pilots

Pilot	Problem found	Decision
Sadda + eSpeak	Real pauses were absorbed into nearby phone durations. Clip 898 also had a suspected false-silence region at 0.66–0.86 s.	Pilot durations were not adopted for full-data training.
Telugu MFA	The 130-entry g2g dictionary produced character/virama labels and several implausible boundaries.	No validated phoneme-versus-character training result.

For clip 898, the suspected “silence” RMS was 0.1778 versus 0.1129 nearby. Listening supported the concern; RMS alone is not a speech detector. Inventory support also does not establish good alignment timing.

Variance adaptor modification

The main change is the position of the pitch, energy and voicing predictors. Character-level prediction assigns one value to an entire character span. Frame-level prediction can represent changes inside that span.

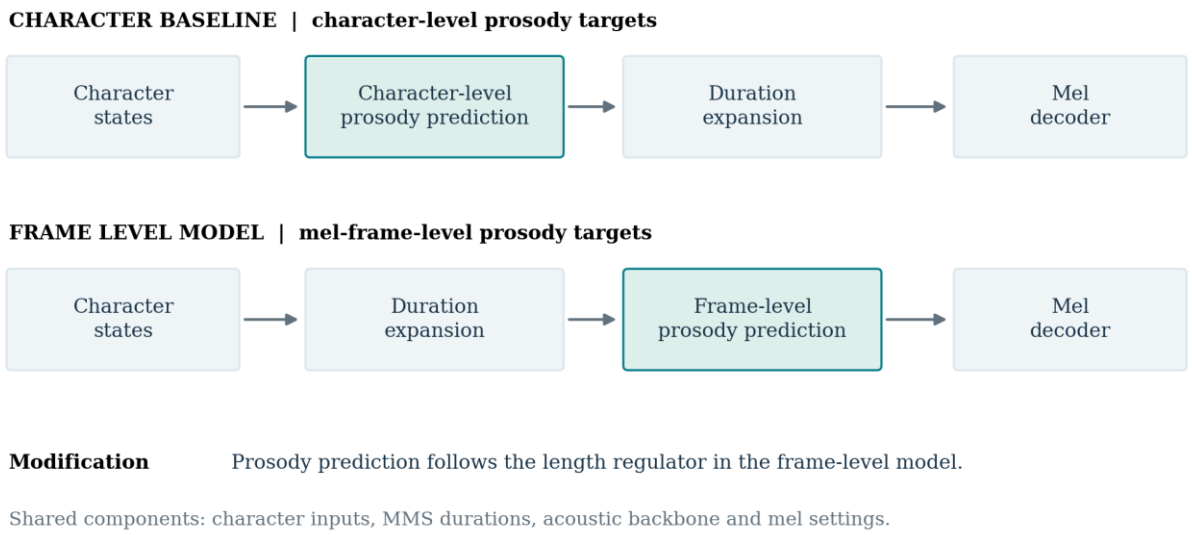


Figure 2. The predictors move after duration expansion. Both variants still use character inputs and MMS duration targets.

Prediction granularity

For a character lasting five mel frames	Character baseline	Frame-level model
Pitch / energy targets	One average value for the span	One value at each frame
Represented variation	Variation between characters	Variation within and between characters
Saved experiment name	character_baseline	frame_prosody

Model implementation. Current checkpoints use PilotFS2, while older FastSpeech2 checkpoints require their original class. The final models retain character inputs; the phoneme-alignment pilots were not used for full-data training.

Model settings and training

Each language and each variant was trained independently, using the same settings and epoch budget within each language.

Shared acoustic model

Component	Setting
Text encoder and decoder	128-dimensional embeddings; sinusoidal positions; 2 encoder and 2 decoder blocks
Transformer blocks	2 attention heads; convolutional feed-forward layers; residual connections; layer normalization; dropout 0.1
Predictors and output	Convolutional duration / pitch / energy / voicing heads; continuous pitch and energy projections; 80-bin mel output
Adaptation from the reference [1,2]	Compact backbone and added voicing head; no speaker embedding, Post-Net or CWT pitch representation

Training settings

Evaluation protocol. Validation mel MAE uses recorded durations, pitch, energy and voicing. Text-only synthesis uses predicted values. An improvement in reference-conditioned mel error does not establish an equivalent improvement in text-only speech quality.

Setting	Value
Optimization	AdamW; fixed learning rate 0.0003; batch size 4; gradient norm clipped to 1.0
Randomness and compute	Seed 42; epoch shuffle seed 42 + epoch; one GPU at a time
Training budget	English: 50 epochs / 40200 updates; Telugu: 60 epochs / 25500 updates
Training objective	Mel L1 + log-duration MSE + voiced-only pitch MSE + energy MSE + voicing BCE. Padding is excluded.

Training results

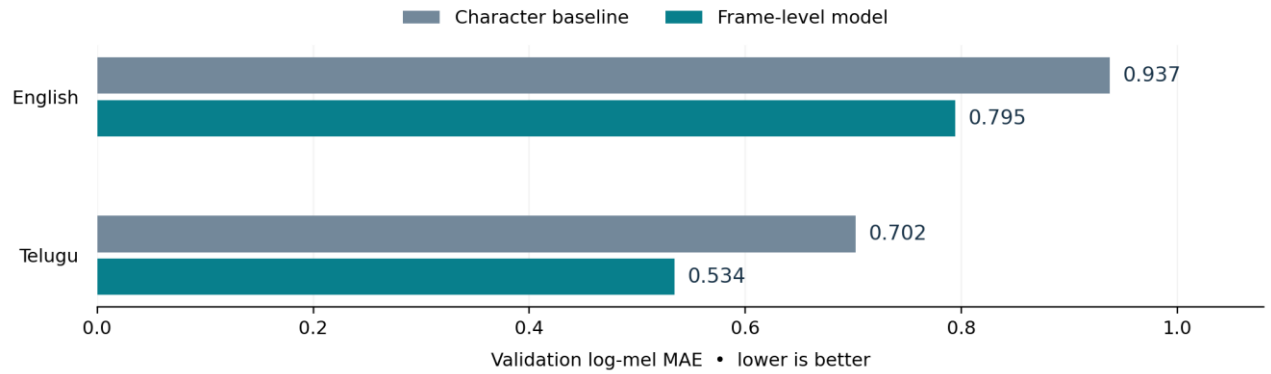


Figure 3. Best reference-conditioned validation MAE. The best checkpoints occur at different epochs; the bars do not compare listening scores.

Language	Variant	Best epoch	Best MAE	Final MAE
English	Character baseline	36	0.937435	0.958347
English	Frame-level	48	0.794563	0.810859
Telugu	Character baseline	58	0.702442	0.706587
Telugu	Frame-level	56	0.534438	0.534669

Quantitative comparison. At the best checkpoints, frame-level validation MAE is lower by 15.2% for English and 23.9% for Telugu. At the final matched epoch budget, the reductions are 15.4% and 24.3%. These percentages describe acoustic prediction error, not measured listening-quality gains.

Mel MAE is weighted by frame count. Total loss is weighted by clip count. The prosody grids differ, so total losses are not directly interchangeable. Selected frame-level checkpoints: English epoch 48 / update 38592; Telugu epoch 56 / update 23800.

Listening evaluation limitation. A systematic sentence-level assessment of the final utterances was not completed. Pronunciation, stress, pauses, noise and overall quality therefore lack formal listening ratings. Informal listening during development identified residual noise in some model outputs.

Output validity. All ten WAVs are finite at 22050 Hz, and waveform length equals mel frames × 256. Saved artifacts include mel arrays, durations, pitch, energy, voicing and metadata. RESULTS.md indexes the audio and all 30 variance plots. Any edited or denoised audio is distinct from the raw model output.

Training and validation convergence

The plots below are preserved from the saved metric CSVs. They show observed training history rather than a smoothed or reconstructed trend.

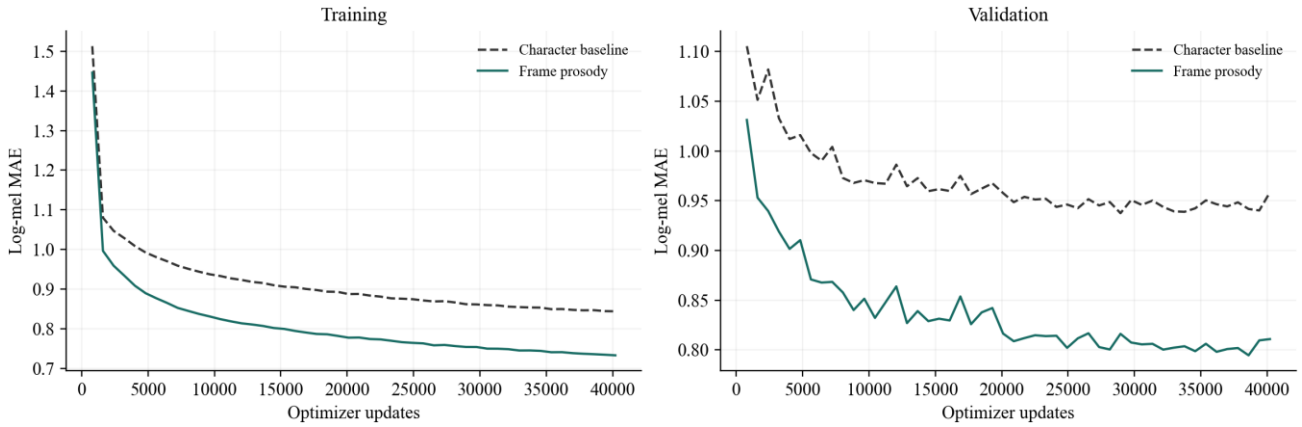


Figure 4. English training and reference-conditioned validation mel MAE across 50 epochs.

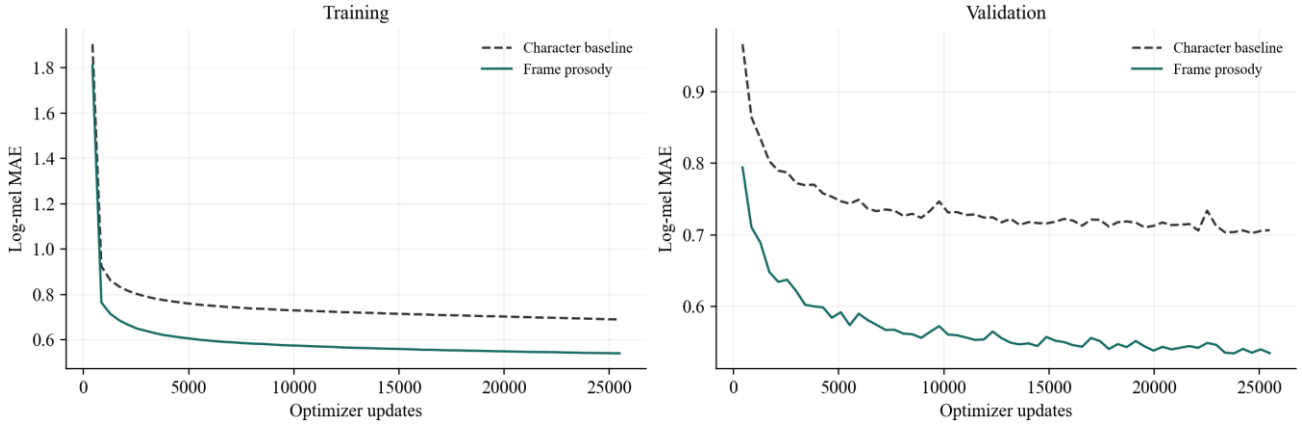


Figure 5. Telugu training and reference-conditioned validation mel MAE across 60 epochs. "Frame prosody" in the legends means frame-level prediction.

Convergence analysis

- Both variants learn: training error decreases. The frame-level curve remains below the character baseline in each language.
- Validation improves more slowly later and fluctuates. The best checkpoint occurs before the final epoch, so best_mel.pt is selected instead of automatically using last.pt.
- There is no clear late validation collapse in these plots. Small increases after the minimum support validation-based selection, but do not establish severe overfitting.

Training and validation conditions. Training loss is measured while weights change and dropout is active; validation uses evaluation mode. A lower validation value than training value does not by itself establish data leakage.

Cross-language interpretation. English and Telugu mel errors are not a direct ranking of speech quality because speakers, recording conditions and content differ between the datasets.

Diagnostic experiments and findings

The Telugu investigation tested preparation, model fitting, alignment and waveform reconstruction. No single test established the complete cause of the remaining noise.

Test	Observed result	Interpretation
Preparation audits	Saved loader tensors matched sampled features; duration sums and padding masks were consistent.	No simple array or padding error was found. Boundary accuracy remained unverified.
Original model continuation	Best validation mel MAE: about 0.7465 after the first 10 epochs; 0.7147 at epoch 27 of the continuation.	More training reduced error, but informal listening still found noise.
Recorded-prosody diagnostic	Training-clip MAE: 91 → 0.6683; 102 → 0.6004; 1038 → 0.6558.	Mel errors remained when recorded prosody was supplied.
One clip and an earlier 16-clip test	One clip could be memorized. The earlier 16-clip MAE fell from 0.4752 at 1000 updates to 0.3193 at 3000.	The model could fit a small set; this did not show generalization.
2500 extra updates from epoch 27	Fixed validation-32: 0.7252 → 0.7189; fixed train-16: 0.6634 → 0.6532.	No clear listening improvement from this short continuation.
Matched 16-clip comparison	At 3000 updates each: baseline 0.3077; frame-level 0.2825, an 8.2% reduction.	An in-sample diagnostic that justified a full-data comparison.
Full-data comparison	Telugu best validation MAE: baseline 0.702442; frame-level 0.534438.	The reference-conditioned advantage also appeared on held-out clips.

Small set memorization and full data generalization

With batch size 4, 3000 updates present about 12000 examples: roughly 750 presentations per clip for a 16-clip set, versus only 7.1 for 1697 clips. The small model was repeatedly seeing the same examples. Its low in-sample loss cannot be treated as a full-data validation result.

Waveform reconstruction with HiFi GAN

The original recording was clear, but its reference-mel reconstruction sounded distorted. This was an important diagnostic: a reference mel bypasses the acoustic model, so that distortion could not be explained only by FastSpeech 2 training.

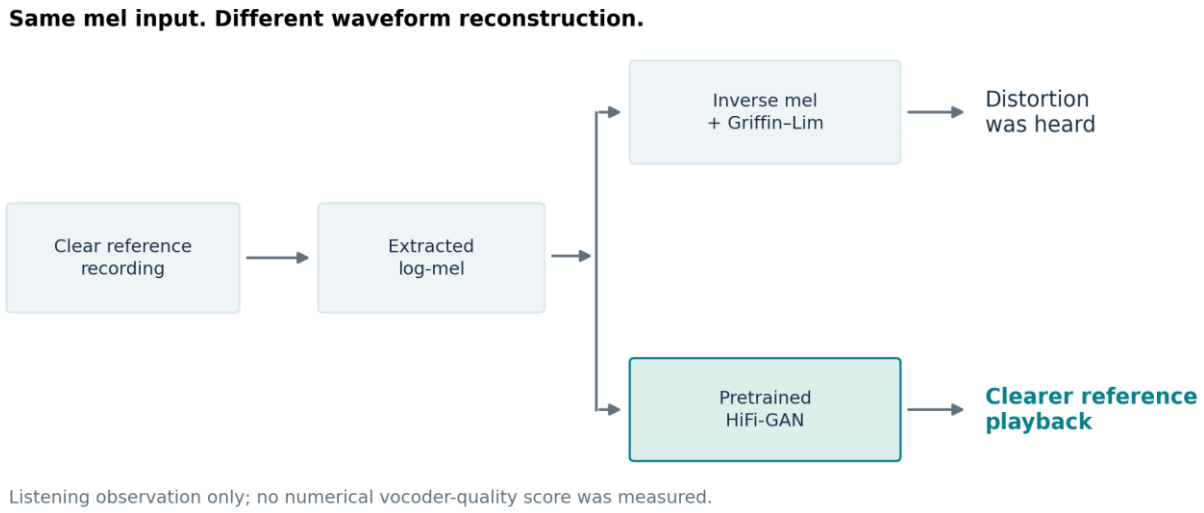


Figure 6. The same reference-mel test isolated a contribution from waveform reconstruction.

Earlier playback	Final playback
exp(log_mel) → inverse mel → Griffin–Lim	Log-mel → pretrained universal HiFi-GAN
Estimates a linear magnitude spectrum and reconstructs phase iteratively; power=1.0.	Uses a learned generator to produce a waveform.
Reference reconstruction sounded distorted in informal listening.	Reference reconstruction sounded much clearer in informal listening.

Scope of the vocoder change

- HiFi-GAN was not trained or fine-tuned in this project. It was used for waveform generation after mel prediction.
- A vocoder change does not alter the MAE of a fixed predicted mel. The lower frame-level mel errors come from the acoustic-model comparison.
- Noise still remained in some model-generated speech. The reconstruction change therefore solved only part of the observed problem.

Vocoder details. Both languages use universal V1 weights g_02500000 [3,4]: 22050 Hz, 80 mel bins, FFT/window 1024, hop 256 and fmax 8000 Hz. Generator commit: 4769534d45265d52a904b850da5a622601885777.

Frontend compatibility. The acoustic frontend is centered, while the original vocoder frontend uses a different padding convention. Exact frontend equivalence remains unverified. No MOS, SNR, ASR-error or numerical vocoder-quality comparison was performed.

Assignment samples and prediction checks

All ten utterances were generated in text-only mode using the best frame-level checkpoints and the same HiFi-GAN. The table summarizes model predictions; it is not a listening assessment.

ID	Audio seconds	Character duration frames	Voiced F0 Hz	Negative energy count	Sentence focus
E1	7.152	2–14	166–258	12	Rare, long word
E2a	4.934	2–24	186–269	0	Noun conduct
E2b	5.097	2–15	148–239	0	Verb conduct
E3	3.437	2–24	180–262	0	Split “rhyth m”
E4	9.509	2–41	126–265	0	Date, time, andaroo
T1	5.956	3–21	151–350	19	Technical expression
T2a	3.471	3–31	163–288	1	Pattu as grip
T2b	3.587	3–30	156–280	9	Pattu as silk
T3	2.508	3–14	149–294	6	Unusual spacing
T4	12.458	3–35	157–300	48	Expanded date/time

Variance adaptor analysis

- Duration: E4 and T4 produce 9.509 s and 12.458 s of audio after date/time expansion. Their maximum predicted character durations are 41 and 35 frames. Integer durations sum to mel length, but timing consistency does not establish natural pauses or accurate speech boundaries.
- Pitch: gaps mark unvoiced or invalid positions. E2a has a voiced F0 range of 186–269 Hz, while E2b has a range of 148–239 Hz. These overlapping ranges do not establish whether the noun/verb stress contrast in conduct was produced correctly.
- Energy: some predictions become negative after de-normalization, which is unphysical for a magnitude norm. T4 has the largest count, 48. The cause of these predictions and their contribution to audible artifacts have not been established.

Normalization of prescribed test sentences

Original assignment text and separate spoken forms are stored in test_sentences.json. E4 and T4 use explicit date/time expansions. Character concatenation removes word boundaries, so the split “rhyth m” becomes the same character sequence as “rhythm”. The two pattu meanings do not automatically require different phoneme sequences.

Reproducibility and software organization

Operation	Implementation	Dependencies and saved state
Inference	Notebook 04: custom-text synthesis. Notebook 03: prescribed assignment sentences.	Uses the matching checkpoint, vocabulary and normalization statistics; training data is not required.
Preprocessing	Notebook 00; English/preprocess.py and Telugu/preprocess.py.	Produces source IDs, split manifests and dataset revision records.
Training	Notebooks 01 and 02; English/train.py and Telugu/train.py.	Uses separate language configurations (config.yaml) and output directories.
Checkpoint resume	last.pt with the corresponding feature and target caches.	The epoch setting is the final target; a completed epoch-60 run resumes through epochs 61–80 when set to 80.

Saved state. best_mel.pt stores the best validation model. last.pt retains the model, optimizer, RNG and partial-epoch state. Atomic saves occur every epoch and every 100 updates. Configuration or source-order mismatches stop resume. Training data is not required for inference.

Validation coverage and reproducibility limitations

The project validation record (VALIDATION.md) reports 6 unit tests, 8 strict checkpoint loads, 20 bitwise-equal CPU prediction comparisons, checks against 10 saved mel outputs, reproduction of one Telugu clip's features, CPU synthesis for both languages, and 5 valid notebooks. These checks cover selected components and saved outputs; they do not establish full-dataset reproduction.

Some original feature archives, final alignment manifests and historical dataset revision information are not distributed. Exact historical resume is therefore limited. A complete fresh Colab run and anonymous access to the checkpoint links were not independently verified.

References and source links

[1] FastSpeech 2 paper
[2] ming024 reference implementation
[3] HiFi-GAN author implementation
[4] Universal V1 weights mirror
[5] Svarah dataset
[6] IndicTTS Telugu dataset
[7] U. Hermjakob, J. May and K. Knight. Out-of-the-box universal romanization tool uroman, 2018.